5.5 Proving Triangle Congruence by SSS

Name _____ Period ____ Date _____

Geometry 5.5 (30-minute plan) · Student sheet

I can…

- use SSS to decide whether two triangles are congruent.
- use HL for right triangles.
- write a proof that uses SSS.

Vocabulary · fill in the missing word

Legs (of a right triangle) — _____

Hypotenuse — the side _____ the right angle

Part 1 · Notes and examples (fill in each note, then work the example)

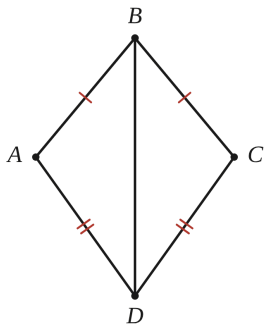
SSS Congruence Theorem: if three _____ of one triangle are congruent to three sides of a second triangle, then the triangles are congruent.

Example 1

One triangle has sides 5, 7, and 9. Another has sides 9, 5, and 7. Are they congruent? Why?

Example 2

Given: $\overline{AB} \cong \overline{CB}$ and $\overline{AD} \cong \overline{CD}$. Prove: $\triangle ABD \cong \triangle CBD$.



Hypotenuse-Leg (HL) Congruence Theorem: if the hypotenuse and a _____ of a right triangle are congruent to the hypotenuse and a leg of a second right triangle, then the triangles are congruent.

Example 3

Two right triangles each have a hypotenuse of 13 and one leg of 5. Are they congruent? Why?

Part 2 · Partner practice

Partner practice

Two right triangles each have a hypotenuse of 10 and a leg of 6. Are they congruent? Why?